

# FEMS Read Only API User Guide

## Overview

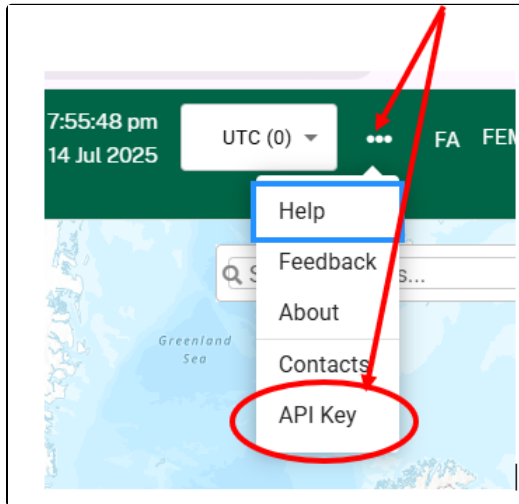
The FEMS Read Only API allows external applications to utilize FEMS data. This guide identifies the instructions for using the FEMS API.

NOTE: Only users with a FEMS API or FEMS Admin role can use the FEMS Read Only API.

## FEMS API Key

Follow the instructions in this section to generate and copy a FEMS API Key for use in accessing the read-only FEMS API:

1. Log into FEMS with a user account that has either the FEMS API or FEMS Admin role.
2. Select the hamburger menu  or the ... menu, if the hamburger menu does not display.
3. Select the **API Key** option.



4. When the FEMS API Key Generate window opens, the user name for the account into which you are logged displays.



8. Save the key in a safe location.

NOTE: Each time you click the Generate API Key button, a new API key will be generated for the logged in user account. When a new FEMS API code is generated, the previous code is no longer valid.

## Using the FEMS API Key

1. Open an API platform like Postman.
2. Select the Basic Authorization option.
3. Enter the User Name associated with the API key.
4. Enter the API key as the password.

The screenshot shows the Postman interface for a request to `https://fems.fs2c.usda.gov/api/ext-climatology/graphql` using the `POST` method. The `Authorization` tab is selected, showing `Basic Auth` as the `Auth Type`. The `Username` field contains `trudi.allison@usda.gov` and the `Password` field is masked with dots. A note states: "The authorization header will be automatically generated when you send the request. Learn more about [Basic Auth](#) authorization."

5. Run the API calls for the following url: <https://fems.fs2c.usda.gov/api/ext-climatology/graphql>

## Example API Calls For Use with FEMS

| API Call     | Query                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Variables                                                                                                                                                                                                                                                                | NOTES                                                                                                                                                                                   |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFDR Catalog | <pre> query NfdrCatalog(   \$stationId: String!   \$fuelModels: [FuelModelTypes]   \$returnAll: Boolean   \$fireDangerModelName: String   \$kdbiParamCatalogName: String   \$catalogName: String   \$page: Int   \$per_page: Int ) {   nfdrCatalog(     returnAll: \$returnAll     stationIds: \$stationId     fuelModels: \$fuelModels     fireDangerModelName: \$fireDangerModelName     kdbiParamCatalogName: \$kdbiParamCatalogName     catalogName: \$catalogName     page: \$page     per_page: \$per_page   ) {     _metadata { </pre> | <p><u>Optional Variables:</u></p> <p>returnAll: Boolean</p> <p>fuelModels: [FuelModelTypes]</p> <p>fireDangerModelName: String</p> <p>kdbiParamCatalogName: String</p> <p>catalogName: String</p> <p>page: Int</p> <p>per_page: Int</p> <p><u>Required Variables</u></p> | <p><b>Acceptable [ FuelModelTypes] values:</b></p> <p>null</p> <p>"V"</p> <p>"W"</p> <p>"X"</p> <p>"Y"</p> <p>"Z"</p> <p><b>Acceptable Boolean values:</b></p> <p>null</p> <p>false</p> |

```
page
per_page
total_count
page_count
}
data {
  fire_danger_model_station_id
  fire_danger_model_id
  fire_danger_model_name
  primary_flg
  fems_station_id
  station_id
  wims_id
  fuel_model
  version
  slope_class
  grass_type
  scm
  mxd
  kbd_i_parameter_catalog_id
  dead_fuel_parameter_catalog_id
  gsi_parameter_catalog_id
  kbd_i_threshold
  kbd_i_initial
  kbd_i_parameter_catalog_name
  catalog_name
  herb_days_avg
  herb_max_gsi
  herb_greenup
  herb_max
  herb_min
  herb_temp_min_max
  herb_temp_min_min
  herb_vpd_min
  herb_vpd_max
  herb_day_length_min
  herb_day_length_max
  herb_vpd_avg_flg
  woody_days_avg
  woody_max_gsi
  woody_greenup
  woody_max
  woody_min
  woody_temp_min_max
  woody_temp_min_min
  woody_vpd_min
  woody_vpd_max
  woody_day_length_min
  woody_day_length_max
  woody_vpd_avg_flg
  herb_precip_total_min
  herb_precip_total_max
  herb_total_precip_days
  herb_use_precip_flg
  woody_precip_total_min
  woody_precip_total_max
  woody_total_precip_days
  woody_use_precip_flg
  herb_calibration
  woody_calibration
  dead_fuel_parameter_catalog_name
  absorption_rate
  stick_radius_1_hr_fuel_moisture
  stick_radius_10_hr_fuel_moisture
  stick_radius_100_hr_fuel_moisture
```

stationIds: String

true

Example GraphQL  
Variables:

```
{
  "returnAll": null,
  "stationId": "353213
, 55522410",
  "fuelModels": null,
  "fireDangerModelName": null,
  "kdbiParameterCatalogName": null,
  "catalogName": null,

  "page": 0,
  "per_page": 100000
}
```

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                          |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | <pre> stick_radius_1000_hr_fuel_moisture modified_time modified_by created_time created_by } } } </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                          |
| NFDR Min Max | <pre> query NfdrMinMax(\$startDate: Date!, \$stationIds: String, \$wrcclds: String, \$fuelModels: String, \$nfdrType: String, \$zoomLevel: Int, \$endDate: Date, \$hasHistoricData: TriState, \$sortBy: NfdrMinMaxSortBy, \$sortOrder: SortOrder, \$page: Int, \$perPage: Int) {   nfdrMinMax(startDate: \$startDate, stationIds: \$stationIds, wrcclds: \$wrcclds, fuelModels: \$fuelModels, nfdrType: \$nfdrType, zoomLevel: \$zoomLevel, endDate: \$endDate, hasHistoricData: \$hasHistoricData, sortBy: \$sortBy, sortOrder: \$sortOrder, page: \$page, per_page: \$perPage) {     _metadata {       page       page_count       per_page       total_count     }     data {       burning_index_max       burning_index_max_time       elevation       energy_release_component_max       energy_release_component_max_time       fuel_model       gsi       herbaceous_lfi_fuel_moisture       hun_hr_tl_fuel_moisture_min       hun_hr_tl_fuel_moisture_min_time       ignition_component_max       ignition_component_max_time       kbdi       latitude       longitude       nfdr_type       one_hr_tl_fuel_moisture_min       one_hr_tl_fuel_moisture_min_time       quality_code       spread_component_max       spread_component_max_time       station_id       station_name       summary_date       ten_hr_tl_fuel_moisture_min       ten_hr_tl_fuel_moisture_min_time       thou_hr_tl_fuel_moisture_min       thou_hr_tl_fuel_moisture_min_time       woody_lfi_fuel_moisture       wrcc_id     }   } } </pre> | <p><b>Optional Variables:</b></p> <p>stationIds: String</p> <p>wrcclds: String</p> <p>fuelModels: String</p> <p>nfdrType: String</p> <p>zoomLevel: Int</p> <p>endDate: Date</p> <p>sortBy: NfdrMinMaxSortBy</p> <p>sortOrder: SortOrder</p> <p>page: Int</p> <p>perPage: Int</p> <p><b>Required Variables:</b></p> <p>startDate: Date!</p> <p>hasHistoricData: TriState</p> <p><b>Example GraphQL Query:</b></p> | <p><b>Acceptable TriState values:</b></p> <p>"ALL"</p> <p>"FALSE"</p> <p>"TRUE"</p> <p><b>Acceptable NfdrMinMaxSortBy Values:</b></p> <p>"station_id"</p> <p>"station_name"</p> <p>"summary_date"</p> <p><b>Acceptable SortOrder Values:</b></p> <p>"asc"</p> <p>"desc"</p> <p><b>Date Format:</b></p> <p>YYYY-MM-DD</p> |

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                       |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <pre>{   "startDate": "2025-07-03",   "stationIds": null,   "wrccls": null,   "fuelModels": null,   "nldrType": null,   "zoomLevel": null,   "endDate": "2025-07-04",   "hasHistoricData": "ALL",   "sortBy": null,   "sortOrder": null,   "page": null,   "perPage": null }</pre> <pre>{   "startDate": "2026-03-12",   "endDate": "2026-04-02",   "stationIds": "102004",   "sortBy": "summary_date",   "sortOrder": "DESC",   "page": 0,   "per_page": 25,   "fuelModel": "Y",   "hasHistoricData": "ALL",    "stationIds": null,   "wrccls": null,    "nldrType": null,   "zoomLevel": null, }</pre> |                                                                                                                                                                                                                                                                                                                                                       |
| NFDRS<br>Obs | <pre>query NfdrsObs(   \$fuelModels: String!   \$stationIds: String   \$wrccls: String   \$nldrType: String   \$zoomLevel: Int   \$startDateRange: Date   \$endDateRange: Date   \$startHour: Int   \$endHour: Int   \$dateTimeFormat: DateTimeFormat   \$hasHistoricData: TriState   \$sortBy: NfdrObsSortBy   \$sortOrder: SortOrder   \$page: Int   \$perPage: Int ){   nfdrsObs(     fuelModels: \$fuelModels     stationIds: \$stationIds     wrccls: \$wrccls     nldrType: \$nldrType     zoomLevel: \$zoomLevel</pre> | <p><u>Optional Variables:</u></p> <p><code>stationIds: String</code></p> <p><code>wrccls: String</code></p> <p><code>nldrType: String</code></p> <p><code>zoomLevel: Int</code></p> <p><code>sortBy: NfdrObsSortBy</code></p> <p><code>sortOrder: SortOrder</code></p> <p><code>page: Int</code></p> <p><code>perPage: Int</code></p> <p><code>startHour: Int</code></p>                                                                                                                                                                                                                                 | <p><b>Acceptable <u>DateTimeFormat</u> values:</b></p> <p>"LocalStationTime"</p> <p>"UTC"</p> <p><b>Acceptable <u>SortOrder</u> Values:</b></p> <p>"asc"</p> <p>"desc"</p> <p><b>Acceptable <u>TriState</u> values:</b></p> <p>"ALL"</p> <p>"FALSE"</p> <p>"TRUE"</p> <p><b>Acceptable <u>NfdrObsSortby</u> values:</b></p> <p>"observation_time"</p> |

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                    |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
|                            | <pre> startDateRange: \$startDateRange endDateRange: \$endDateRange startHour: \$startHour endHour: \$endHour dateTimeFormat: \$dateTimeFormat hasHistoricData: \$hasHistoricData sortBy: \$sortBy sortOrder: \$sortOrder page: \$page per_page: \$perPage ){   _metadata {     page     per_page     total_count     page_count   }   data {     station_name     station_id     wrcc_id     latitude     longitude     elevation     observation_time     observation_time_1st     display_hour     display_hour_1st     masked_observation_time     nldr_date     nldr_time     nldr_type     fuel_model     fuel_model_version     kbd_i     one_hr_tl_fuel_moisture     ten_hr_tl_fuel_moisture     hun_hr_tl_fuel_moisture     thou_hr_tl_fuel_moisture     ignition_component     spread_component     energy_release_component     burning_index     herbaceous_lfi_fuel_moisture     woody_lfi_fuel_moisture     gsi     quality_code   } } } </pre> | <pre> endHour: Int dateTimeFormat: DateTimeFormat fuelModels: String! startDateRange: Date endDateRange: Date hasHistoricData: TriState  Example GraphQL Query: {   "startDateRange": "2025-07-03",   "endDateRange": "2025-07-04",   "hasHistoricData": "ALL",   "sortBy": "",   "fuelModels": "Y",   "wrccIds": null,   "stationIds": null,   "nldrType": null,   "zoomLevel": null,   "startHour": null,   "endHour": null,   "dateTimeFormat": "LocalStationTime",   "sortOrder": null,   "page": null,   "perPage": null, } </pre> | <p>"station_id"</p> <p><b>Acceptable Date Format:</b></p> <p>"YYYY-MM-DD"</p> <p><b>Acceptable TimeHour Format</b></p> <p>"HH"</p> |
| Get Percentile Avg Min Max | <pre> query GetPercentileAvgMinMax(   \$stationId: String!   \$fuelModel: FuelModelTypes!   \$startYear: ClimatologyYear   \$endYear: ClimatologyYear   \$startMonthDay: ClimatologyMonthDay   \$endMonthDay: ClimatologyMonthDay   \$startHour: TimeHour   \$endHour: TimeHour   \$dateTimeFormat: DateTimeFormat </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p><u>Optional Variables:</u></p> <p>startYear: ClimatologyYear</p> <p>endYear: ClimatologyYear</p> <p>startMonthDay: ClimatologyMonthDay</p>                                                                                                                                                                                                                                                                                                                                                                                           | <p><b>Acceptable FuelModelTypes! values:</b></p> <p>null</p> <p>"V"</p> <p>"W"</p> <p>"X"</p>                                      |

```

){
  percentileAvgMinMax(
    stationIds: $stationId
    fuelModel: $fuelModel
    climatology: {
      startYear: $startYear
      endYear: $endYear
      startMonthDay: $startMonthDay
      endMonthDay: $endMonthDay
      startHour: $startHour
      endHour: $endHour
      dateTimeFormat: $dateTimeFormat
    }
    page: 0
    per_page: 300000
  ){
    _metadata {
      page
      per_page
      total_count
      page_count
    }
    data {
      station_id
      day
      energy_release_component_max {
        min
        max
        avg
        stddev
      }
      spread_component_max {
        min
        max
        avg
        stddev
      }
      ignition_component_max {
        min
        max
        avg
        stddev
      }
      burning_index_max {
        min
        max
        avg
        stddev
      }
      one_hr_tl_fuel_moisture_min {
        min
        max
        avg
        stddev
      }
      ten_hr_tl_fuel_moisture_min {
        min
        max
        avg
        stddev
      }
      hun_hr_tl_fuel_moisture_min {
        min
        max
        avg

```

```

endMonthDay: ClimateMonthDay
endHour: TimeHour
startHour: TimeHour
dateTimeFormat: DateTimeFormat

Required Variables:

fuelModel: FuelModelTypes!
stationId: String!

```

Example GraphQL Query:

```

{
  "stationId": "45220
, 55522410",
  "fuelModel": "Y",
  "startYear": "2005",

  "endYear": "2022",

  "startMonthDay": "
01-01",
  "endMonthDay": "1
2-31",
  "startHour": "0",
  "endHour": "24",
  "dateTimeFormat":
"LocalStationTime"
}

```

"Y"

"Z"

Acceptable ClimatologyYear format:

"YYYY"

Acceptable ClimatologyMonthDay format:

"MM-DD"

Acceptable TimeHour Format

"HH"



```
    stddev
  }
  thou_hr_tl_fuel_moisture_min {
    min
    max
    avg
    stddev
  }
  herbaceous_lfi_fuel_moisture_max {
    min
    max
    avg
    stddev
  }
  woody_lfi_fuel_moisture_max {
    min
    max
    avg
    stddev
  }
  kbdi_max {
    min
    max
    avg
    stddev
  }
  gsi_max {
    min
    max
    avg
    stddev
  }
  burning_index {
    min
    max
    avg
    stddev
  }
  energy_release_component {
    min
    max
    avg
    stddev
  }
  gsi {
    min
    max
    avg
    stddev
  }
  herbaceous_lfi_fuel_moisture {
    min
    max
    avg
    stddev
  }
  hun_hr_tl_fuel_moisture {
    min
    max
    avg
    stddev
  }
  ignition_component {
    min
    max
```

```
    avg
    stddev
  }
  kbd {
    min
    max
    avg
    stddev
  }
  one_hr_tl_fuel_moisture {
    min
    max
    avg
    stddev
  }
  spread_component {
    min
    max
    avg
    stddev
  }
  ten_hr_tl_fuel_moisture {
    min
    max
    avg
    stddev
  }
  thou_hr_tl_fuel_moisture {
    min
    max
    avg
    stddev
  }
  woody_lfi_fuel_moisture {
    min
    max
    avg
    stddev
  }
}

vapor_press_def_max{
  min
  max
  avg
  stddev
}

vapor_press_def_min{
  min
  max
  avg
  stddev
}
}
```

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                         |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Percentile Levels | <pre> query GetPercentileLevels(   \$stationId: String!   \$fuelModel: FuelModelTypes!   \$startYear: ClimatologyYear   \$endYear: ClimatologyYear   \$startMonthDay: ClimatologyMonthDay   \$endMonthDay: ClimatologyMonthDay   \$startHour: TimeHour   \$endHour: TimeHour   \$percentileLevels: String   \$dateTimeFormat: DateTimeFormat ){   percentileLevels(     stationIds: \$stationId     fuelModel: \$fuelModel     climatology: {       startYear: \$startYear       endYear: \$endYear       startMonthDay: \$startMonthDay       endMonthDay: \$endMonthDay       startHour: \$startHour       endHour: \$endHour       dateTimeFormat: \$dateTimeFormat     }     percentileLevels: \$percentileLevels     page: 0     per_page: 300000   ){     _metadata {       page       per_page       total_count       page_count     }     data {       station_id       kbdi_max       one_hr_tl_fuel_moisture_min       ten_hr_tl_fuel_moisture_min       hun_hr_tl_fuel_moisture_min       thou_hr_tl_fuel_moisture_min       ignition_component_max       spread_component_max       energy_release_component_max       woody_lfi_fuel_moisture_max       herbaceous_lfi_fuel_moisture_max       burning_index_max        vapor_press_def_max        vapor_press_def_min     }   } }</pre> | <p><u>Optional Variables:</u></p> <p>startYear: ClimatologyYear</p> <p>endYear: ClimatologyYear</p> <p>startMonthDay: ClimatologyMonthDay</p> <p>endMonthDay: ClimatologyMonthDay</p> <p>endHour: TimeHour</p> <p>startHour: TimeHour</p> <p>dateTimeFormat: DateTimeFormat</p> <p><u>Required Variables:</u></p> <p>fuelModel: FuelModelTypes!</p> <p>stationId: String!</p> <p>percentileLevels: String</p> <p><u>Example GraphQL Variable:</u></p> <pre>{   "stationId": "45220,55522410",   "fuelModel": "Y",   "startYear": "2005",    "endYear": "2022",    "startMonthDay": "01-01",   "endMonthDay": "12-31",   "startHour": "0",   "endHour": "24",   "dateTimeFormat": "LocalStationTime",   "percentileLevels": "90,97,3,10" }</pre> | <p><b>Acceptable <u>FuelModelTypes!</u> values:</b></p> <p>null</p> <p>"V"</p> <p>"W"</p> <p>"X"</p> <p>"Y"</p> <p>"Z"</p> <p><b>Acceptable <u>ClimatologyYear</u> format:</b></p> <p>"YYYY"</p> <p><b>Acceptable <u>ClimatologyMonthDay</u> format:</b></p> <p>"MM-DD"</p> <p><b>Acceptable <u>TimeHour</u> Format</b></p> <p>"HH"</p> |
| Station Metadata  | <pre> query StationMetaData(   \$returnAll: Boolean   \$zoomLevel: Int   \$stationIds: String   \$networkName: String   \$stateId: String</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p><u>Optional Variables:</u></p> <p>returnAll: Boolean</p> <p>zoomLevel: Int</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p><b>Acceptable <u>TriState</u> values:</b></p> <p>"ALL"</p>                                                                                                                                                                                                                                                                           |

```
$agency: String
$stationType: String
$returnList: Boolean
$sortBy: StationMetadataSortBy
$sortOrder: SortOrder
$hasHistoricData: TriState
$page: Int
$perPage: Int
){
  stationMetaData(
    returnAll: $returnAll
    zoomLevel: $zoomLevel
    stationIds: $stationIds
    networkName: $networkName
    stateId: $stateId
    agency: $agency
    stationType: $stationType
    returnList: $returnList
    hasHistoricData: $hasHistoricData
    sortBy: $sortBy
    sortOrder: $sortOrder
    page: $page
    per_page: $perPage
  ){
    _metadata {
      page
      per_page
      total_count
      page_count
    }
    data {
      fems_station_id
      station_id
      wims_id
      nesdis_id
      wrcc_id
      station_name
      state
      county_name
      latitude
      longitude
      elevation
      slope
      aspect
      aspect_direction
      aspect_degree
      kbdi_threshold
      init_kbdi
      avg_annual_precip
      annual_maintenance_date_time
      period_record_start
      period_record_stop
      station_status
      class
      network_id
      network_name
      agency
      region
      unit
      sub_unit
      ownership_type
      goes
      tx_frequency
      obs_frequency
      site_description
    }
  }
}
```

|                                        |                                                          |
|----------------------------------------|----------------------------------------------------------|
| stationIds: String                     | "FALSE"                                                  |
| networkName: String                    | "TRUE"                                                   |
| stateId: String                        |                                                          |
| agency: String                         | Acceptable <a href="#">StationMetadataSortBy</a> Values: |
| stationType: String                    | "station_id"                                             |
| returnList: Boolean                    | "station_name"                                           |
| sortBy: StationMetadataSortBy          | "network_name"                                           |
| sortOrder: SortOrder                   | "agency"                                                 |
| page: Int                              | "state"                                                  |
| perPage: Int                           | Acceptable <a href="#">SortOrder</a> Values:             |
|                                        | "asc"                                                    |
|                                        | "desc"                                                   |
| <a href="#">Required Variables</a>     |                                                          |
| hasHistoricData: TriState              |                                                          |
| <a href="#">Example GraphQL Query:</a> |                                                          |
| {                                      |                                                          |
| "returnAll": true,                     |                                                          |
| "zoomLevel": null,                     |                                                          |
| "stationIds": "                        |                                                          |
| 353213, 55522410",                     |                                                          |
| "networkName": null                    |                                                          |
| ,                                      |                                                          |
| "stateId": null,                       |                                                          |
| "agency": null,                        |                                                          |
| "stationType": null,                   |                                                          |
| "returnList": null,                    |                                                          |
| "hasHistoricData": "                   |                                                          |
| ALL",                                  |                                                          |
| "sortBy": null,                        |                                                          |
| "sortOrder": null,                     |                                                          |
| "page": null,                          |                                                          |
| "perPage": null                        |                                                          |
| }                                      |                                                          |

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                  |  |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|                | <pre> maintenance_standard reg_scheduled_observation_time time_zone time_zone_offset zoom_level transmit_time modified_time modified_by created_time created_by fuel_models {   fuel_model   version   primary_flg   slopeclass   grasstype   scm   extmoi   last_modified_date } first_observation last_observation site observing_agency gsi_parameter_catalog_id station_type nldr_visibility station_metadata_visible has_historic_data weather_daily_visible nldr_daily_visible weather_hourly_visible nldr_hourly_visible weather_daily_download weather_hourly_download nldr_daily_download nldr_hourly_download produce_nldr ingest_observation  vpd_min  vpd_max  vpd_max_percentile  vpd_min_percentile } } </pre> | <pre> {   "returnAll": true,   "zoomLevel": null,   "stationIds": null,   "networkName": null ,   "stateId": null,   "agency": null,   "stationType": null,   "returnList": null,   "hasHistoricData": " ALL",   "sortBy": null,   "sortOrder": null,   "page": null,   "perPage": null } </pre> |  |
| Weather<br>Obs | <pre> query WeatherObs(   \$startDateTimeRange: DateTime!   \$endDateTimeRange: DateTime!   \$stationIds: String   \$wrccls: String   \$attribute: String   \$timeZone: String   \$zoomLevel: Int   \$hasHistoricData: TriState   \$sortBy: WxObsSortBy   \$sortOrder: SortOrder </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p><u>Optional Variables:</u></p> <pre> wrccls: String  zoomLevel: Int  stationIds: String  attribute: String  timeZone: String </pre>                                                                                                                                                           |  |

```
$page: Int
$perPage: Int
){
  weatherObs(
    startDateTimeRange: $startDateTimeRange
    endDateTimeRange: $endDateTimeRange
    stationIds: $stationIds
    wrcclds: $wrcclds
    attribute: $attribute
    timeZone: $timeZone
    zoomLevel: $zoomLevel
    hasHistoricData: $hasHistoricData
    sortBy: $sortBy
    sortOrder: $sortOrder
    page: $page
    per_page: $perPage
  ){
    _metadata {
      page
      per_page
      total_count
      page_count
    }
    data {
      station_id
      wrcc_id
      station_name
      latitude
      longitude
      elevation
      zoom_level
      station_type
      observation_time
      observation_time_lst
      display_hour
      display_hour_lst
      masked_observation_time
      display_date
      temperature
      relative_humidity
      hourly_precip
      wind_speed
      wind_direction
      peak_gust_speed
      peak_gust_dir
      sol_rad
      snow_flag
      observation_type
      hex
      t_flag
      rh_flag
      pcp_flag
      ws_flag
      wa_flag
      sr_flag
      gs_flag
      ga_flag

      vapor_press_def
    }
  }
}
```

hasHistoricData: TriS  
tate

sortBy: WxObsSortBy

sortOrder: SortOrder

page: Int

perPage: Int

Required Variables

startDateTimeRange:  
DateTime!

endDateTimeRange:  
DateTime!

Example GraphQL  
Query:

```
{
  "startDateTimeRange": "2026-03-26T16:30:00Z",
  "endDateTimeRange": "2026-03-26T17:29:59Z",
  "perPage": null,
  "page": null,
  "sortOrder": null,
  "sortBy": null,
  "hasHistoricData": null,
  "zoomLevel": null,
  "timeZone": null,
  "attribute": null,
  "wrcclds": null,
  "stationIds": null
}
```

```
{
  "stationId": "336001, 55522410",
  "zoomLevel": 5,
  "startDate": "2025-07-14T17:30:00Z",
  "endDate": "2025-07-20T18:29:59Z",
  "hasHistoricData": "ALL"
}
```

1. To include local station time for this query:  
Run the Station Metadata query and request the Time Zone and Time Zone offset:

time\_zone  
time\_zone\_offset

2. Add the offset to the weatherObs variables:  
{ "stationId": "204201", "startDateTimeRange": "2025-07-29T09:33:22-06:00", "endDateTimeRange": "2025-07-29T11:03:22-06:00" }

3. Local station time displays in the response as display\_hour\_lst. UTC time displays in the response as display\_hour:

"observation\_time": "2025-07-14T17:50:24.000Z",  
"observation\_time\_lst": "2025-07-14T12:50:24-05:00",

**Acceptable WxObsSortBy Values:**

"station\_id"  
"station\_name"  
"observation\_time"  
"temperature\_min"  
"temperature"  
"relative\_humidity"  
"hourly\_precip"  
"wind\_speed"  
"wind\_direction"  
"peak\_gust\_speed"  
"peak\_gust\_direction"  
"sol\_rad"  
"vapor\_press\_def"

**Acceptable SortOrder Values:**

"asc"  
"desc"

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                   |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Weighted Winds | <pre>query WeightedWinds(   \$stationIds: String!   \$startYear: ClimatologyYear   \$endYear: ClimatologyYear   \$startMonthDay: ClimatologyMonthDay   \$endMonthDay: ClimatologyMonthDay   \$startHour: TimeHour   \$endHour: TimeHour   \$dateTimeFormat: DateTimeFormat   \$windType: WindType ) {   weightedWinds(     stationIds: \$stationIds     climatology: {       startYear: \$startYear       endYear: \$endYear       startMonthDay: \$startMonthDay       endMonthDay: \$endMonthDay       startHour: \$startHour       endHour: \$endHour       dateTimeFormat: \$dateTimeFormat     }     windType: \$windType   ) {     station_id     calmThreshold     calmWeight     validRecords     totalRows     windType     table   } }</pre> | <p><u>Optional Variables:</u></p> <p><b>startYear:</b> ClimatologyYear</p> <p><b>endYear:</b> ClimatologyYear</p> <p><b>startMonthDay:</b> ClimatologyMonthDay</p> <p><b>endMonthDay:</b> ClimatologyMonthDay</p> <p><b>endHour:</b> TimeHour</p> <p><b>startHour:</b> TimeHour</p> <p><b>dateTimeFormat:</b> DateTimeFormat</p> <p><b>windType:</b> WindType</p> <p><u>Required Variables</u></p> <p><b>stationIds:</b> String!</p> <p><u>Example GraphQL Query:</u></p> <pre>{   "stationIds": "102004",   "windType": "avg",   "startYear": "2005",   "endYear": "2022",   "startMonthDay": "01-01",   "endMonthDay": "12-31",   "startHour": "0",   "endHour": "24",   "dateTimeFormat": "LocalStationTime" }</pre> | <p><b>Acceptable ClimatologyYear format:</b></p> <p>"YYYY"</p> <p><b>Acceptable ClimatologyMonthDay format:</b></p> <p>"MM-DD"</p> <p><b>Acceptable TimeHour Format</b></p> <p>"HH"</p> <p><b>Acceptable windType Values:</b></p> <p>"both"</p> <p>"avg"</p> <p>"gust"</p> <p><b>Acceptable DateTimeFormat values:</b></p> <p>"LocalStationTime"</p> <p>"UTC"</p> |
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|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| wxMinMax | <pre>query WxMinMax(   \$startDate: Date!   \$stationIds: String   \$wrcclDs: String   \$endDate: Date   \$zoomLevel: Int   \$hasHistoricData: TriState   \$sortBy: WxMinMaxSortBy   \$sortOrder: SortOrder   \$page: Int   \$perPage: Int ) {   wxMinMax(     startDate: \$startDate     stationIds: \$stationIds     wrcclDs: \$wrcclDs     endDate: \$endDate     zoomLevel: \$zoomLevel     hasHistoricData: \$hasHistoricData     sortBy: \$sortBy     sortOrder: \$sortOrder     page: \$page     per_page: \$perPage   ) {     _metadata {       page       per_page       total_count       page_count     }     data {       station_name       station_id       wrcc_id       latitude       longitude       elevation       summary_date       observation_type       temperature_min       temperature_max       relative_humidity_min       relative_humidity_max       wind_speed_min       wind_speed_max       peak_gust_speed_min       peak_gust_speed_max       peak_wind_gust_direction       peak_wind_gust_time       solar_radiation_max       daily_precipitation_total       snow_flag        vapor_press_def_min       vapor_ress_def_max     }   } }</pre> | <p><u>Optional Variables:</u></p> <p>wrcclDs: String</p> <p>zoomLevel: Int</p> <p>stationIds: String</p> <p>sortBy: WxMinMaxSortBy</p> <p>sortOrder: SortOrder</p> <p>page: Int</p> <p>perPage: Int</p> <p>endDate: Date</p> <p><u>Required Variables</u></p> <p>startDate: Date!</p> <p>hasHistoricData: TriState</p> <p><u>Example GraphQL Query:</u></p> <pre>{   "startDate": "2026-03-25",   "stationIds": null,   "wrcclDs": null,   "endDate": null,   "zoomLevel": null,   "hasHistoricData": "ALL",   "sortBy": null,   "sortOrder": null,   "page": null,   "perPage": null }</pre> | <p><b>Acceptable WxMinMaxSortBy Values:</b></p> <p>"station_id"</p> <p>"station_name"</p> <p>"summary_date"</p> <p>"temperature_min"</p> <p>"temperature_max"</p> <p>"relative_humidity_min"</p> <p>"relative_humidity_max"</p> <p>"daily_precipitation_total"</p> <p>"wind_speed_min"</p> <p>"wind_speed_max"</p> <p>"peak_gust_speed_min"</p> <p>"peak_gust_speed_max"</p> <p>"peak_wind_gust_direction"</p> <p>"solar_radiation_max"</p> <p>"vapor_press_def_min"</p> <p>"vapor_press_def_max"</p> <p><b>Acceptable SortOrder Values:</b></p> <p>"asc"</p> <p>"desc"</p> <p><b>Acceptable TriState values:</b></p> <p>"ALL"</p> <p>"FALSE"</p> <p>"TRUE"</p> <p><b>Date Format:</b></p> <p>YYYY-MM-DD</p> |
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